JAVASCRIPT | ASSIGNMENT 6

Arrays, Objects, and Core Methods

Task: Create an array of product objects (with name and price properties). Use forEach() to print each product, filter() to find items under a certain price, and map() to create an array of formatted name-price strings.

CODE

```
// Assignment 6
const products = [
  { name: "Laptop", price: 800 },
  { name: "Mouse", price: 25 },
  { name: "Keyboard", price: 45 }
];

// forEach: print each product
console.log("-- All Products (forEach) --");
products.forEach(function (product) {
  console.log("Item: " + product.name + " - Price: $" + product.price);
});

// filter: items under $50
console.log("-- Filtered Products (Price < $50) --");
const cheap = products.filter(product => product.price < 50);
console.log(cheap);

// map: formatted name-price strings
console.log("-- Mapped Product Tags (map) --");
const tags = products.map(product => product.name.toUpperCase() + " costs $" + product.price);
console.log(tags);
```

OUTPUT

```
-- All Products (forEach) --
Item: Laptop - Price: $800
Item: Mouse - Price: $25
Item: Keyboard - Price: $45
-- Filtered Products (Price < $50) --
[ { name: 'Mouse', price: 25 }, { name: 'Keyboard', price: 45 } ]
-- Mapped Product Tags (map) --
[ 'LAPTOP costs $800', 'MOUSE costs $25', 'KEYBOARD costs $45' ]
```

NOTES

- forEach runs a function on each item and returns nothing.
- filter returns a new array with only the items that pass the test.
- map returns a new array with each item transformed. The original array is unchanged.
- Node.js prints strings inside arrays with single quotes. This is only a display difference.